

Business Case:

The Stellar Nexus – A Galactica Odyssey

Author: Rishabh Ray

Student ID: 34525416

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1. Business Justification

1.1 Background

Portfolio Goals: NexaForge Studios aims to increase revenue by 17% over the next three years and significantly improve the player's Day 1, Day 7, and Three-Month retention period over the next three years by expanding their portfolio with innovative gaming projects.

Client Context: NexaForge is a market leader in the Double-A gaming industry, and it balances innovation and accessibility to reach a wider audience on all gaming platforms. Their current annual revenue is \$120 million, with an expected growth of 17% over the next 3 years.

Project Alignment: The stellar Nexus, a Sci-fi exploration-based adventure game, aims to offer a non-linear narrative and a procedurally generated gaming environment to the audience that aligns with NexaForge's strategic goals and enhances the player engagement and retention factor. [5]

KPI justification: "Stellar Nexus" will be designed to achieve the key portfolio KPIs, including

- Day 1 Retention (targeting an increase from 30% to 45%)
- Day 7 Retention (from 12% to 30%)
- Three-Month Retention (from 8% to 16%).

These improvements are expected as the game's dynamic content and player-driven mechanics encourage long-term engagement. [4]

Value Fit: "Stellar Nexus" is based on compelling narration, Emotional story-telling and hand-painted environmental elements, where players encounter dynamic events, forge alliances, and influence the galactic balance using diplomacy and personal instinct, "Stellar Nexus" will reinforce the studio's reputation for delivering innovative and accessible gaming experiences.

1.2 Current Situation and Problem/Opportunity Statement

Problem: The lack of an RPG (role-playing game) particularly in the Sci-fi exploration genre in the offering by NexaForge has created a gap in their offering and hence the introduction of "Stellar Nexus" can be leveraged to attract new players and enhance the player retention with the current existing audience.

Opportunity: Addressing this gap of an RPG in the galactic genre with "Stellar Nexus" presents a significant opportunity for NexaForge Studios to capture a new gaming market segment. The notion of players deciding their roles and paths in the game will provide success and could lead to higher revenue and stronger player loyalty, directly supporting NexaForge's strategic goals. [1]

Value Delivery: It provides value to NexaForge's already shining reputation by setting a new standard for Role-Playing in the Double-A gaming market. It offers a premium offering for the audience at a more accessible rate.

Consideration: The impact of failing to address this opportunity could result in stagnation or a decline in market share as competitors continue to innovate.

1.2.1 Analysis of Options

Option 1: Remake of an old successful game by NexaForge Studio

Brief Outline: Develop a new variation of a popular old game previously made by NexaForge Studios

Problem Fit: Addresses the need for deeper engagement through nostalgia, which is a key driver of player retention. It helps build on the previous success of an earlier game.

Financial Implications: Moderate cost with high potential for player retention.

Conclusion: A balanced remake with essential requirements could attract a broad market appeal which includes old and new audiences but analysing the correct balance between old and new feature sets can be tough to implement. [7]

Option 2: Develop a mobile-based addictive strategy game

Brief Outline: Prioritize the creation of a quick-to-play game for the mobile-based audience

Problem Fit: Tackles the demand for audience retention and long-term engagement through quick-to-play content.

Financial Implications: low cost, low content diversity.

Conclusion: Provides an expansive mobile-based audience but risks losing out on the major chunk of PC and console-based gamers which will hinder the future growth of the Studio. (n.d.). Game Developer | Game Industry News, Deep Dives, and Developer Blogs.

Option 3: A Strategy based game based on story-telling (Recommended)

Brief Outline: Combine non-linear storytelling with procedural generation and diplomacy mechanics.

Problem Fit: Balances engagement, content diversity, and narrative depth.

Financial Implications: Balanced cost, optimal engagement.

Conclusion and Recommendation: This approach offers a balance of innovation, feasibility, and long-term value, which includes narrative depth and gameplay diversity.

1.3 Project and Deliverable Description

Deliverables:

1. A fully developed game titled "Stellar Nexus" featuring - Gradually generated galaxies, non-linear exploration, artistic visuals and a dynamic narrative system.
2. Post-launch support plan including bug fixes and content updates.

Characteristics/Attributes:

1. **Procedural Generation:** This can be measured by the game time of each player and the number of galaxies created by each player
2. **Non-Linear Narrative:** This will be measured by the number of branches created by every player.
3. **Diplomacy Mechanics:** It can be measured by the players' role the end goal reached by the player in the game and the outcome at the end of the gameplay

1.4 Ethical considerations and implications

1. **Promoting Inclusivity and Diversity:** We should understand the transformative power of inclusivity and strive to represent diverse voices, cultures, and perspectives.[\[2\]](#)
2. **Player Well-being and Engagement:** We need to embrace strategies that encourage healthy gaming habits, discourage excessive playtime, and advocate for regular breaks.[\[2\]](#)
3. **Sensitivity to Content:** Consideration of age-appropriate content, about violence and light-sensitive content.[\[2\]](#)

2. Schedule Estimate, Budget Estimate & Financial Analysis

2.1 Schedule Estimate

Timeline: 12 months from project initiation to go live.

Milestones: [\[6\]](#)

- **Milestone 1:** Completion of the game design document (Month 2) - Establishes project scope and aligns the development team with the approval of NexaForge Studio.
- **Milestone 2:** Complete a playable prototype (Month 4) - Create a playable level in the game for review and get approval on the foundational game mechanics and environment setting.
- **Milestone 3:** Internal testing phase (Month 6) - This is a key phase to identifying bugs and determining the gameplay improvements that can be implemented.
- **Milestone 4:** Releasing a Gold Candidate (Month 10) - a version of the game that would be signed off and used for mass duplication of the final product.
- **Milestone 5:** Game launch (Month 12) - Final delivery of “The Stellar Nexus” to the client and release on all the gaming platforms.

Feasibility:

The requirement of non-linear and procedurally generated content in-game can take more time and it requires necessitating overlapping of different phases of the project to complete the project according to the timeline, also the need for artistic visuals needs to be properly categorised in the pre-game design as designing every element to detail will take excess time and storage.

2.2 Preliminary estimate of the costs involved

The budget for the implementation of the gaming project was mainly allocated to staff salaries, hardware and software procurement, and subsequent maintenance. Despite the initial costs, the investment is projected to yield a positive return, with a calculated **NPV of \$767,297 AUD** and an **ROI of 42.27%** (refer to Exhibit A) over two years. The overall budget of the project is **\$1,071,850 AUD**.

The development costs are broken down as follows:

1. **Staffing Costs:**
 - The staffing costs for roles such as Project Manager, Game Developer, Game Designer, Animator, UI/UX Designer, Data Scientist, and QA/Test Analyst amounted to a total of **\$766,850 AUD**.
 - This represents **71.54%** of the total budget (Refer to the Assumptions in the provided document).
2. **Hardware and Software Costs:**
 - The total costs for development tools and licenses, hardware upgrades, testing hardware devices, and cloud hosting services were **\$105,000 AUD**.
 - This constitutes **9.80%** of the total budget (Refer to Exhibit B).
3. **Contingency Fund:**
 - An allocation of **\$200,000 AUD** was set aside for unforeseen expenses, accounting for **18.66%** of the budget.
4. **Maintenance Costs:**
 - The estimated maintenance costs, which include minor updates, bug fixes, major updates, and additional content over two years, amount to **\$350,000 AUD**.

Financial Metrics:

- **Total Costs (Cash Outflows):** \$1,815,348 AUD
- **Total Benefits (Cash Inflows):** \$2,582,645 AUD
- **Net Present Value (NPV):** \$767,297 AUD
- **Return on Investment (ROI):** 42.27%

3. Development Approach

Approach:

Agile Development with iterative releases and continuous feedback from the client.

Justification:

- **Project Type:** The iterative agile approach allows flexibility in updating features based on the stakeholder input.
- **Stakeholder Feedback:** Regular sprint reviews ensure that developers, the stakeholders and player feedback (during alpha testing) are incorporated throughout development.
- **Feasibility:** Agile is a well-suited method for managing complex projects like "The Stellar Nexus," as ongoing adjustments are crucial for the success of this project.

4. Risk and Quality

4.1 Uncertainty & Risk Analysis

Risk 1: Complexity of the Procedural Generation Algorithm

- **Likelihood:** Medium
- **Impact:** High – May affect game diversity and player retention.
- **Potential Response:** Iterative testing and refinement; allocate additional resources to the development team if necessary.

Risk 2: Competition with other RPG

- **Likelihood:** High
- **Impact:** Medium– Risk of similar titles releasing together.
- **Potential Response:** Distinguish the game through unique features and strong marketing to emphasise its unique selling points.

4.2 Quality Metrics and Measurement

KPI 1: Day 7 Retention Rate

- **How Measured:** Tracked through in-game analytics

KPI 2: Player Engagement

- **How Measured:** Average playtime per user monitored through backend systems.

Financial Exhibits

- NPV Calculation:

	A	B	C	D	E
1	Discount rate	10%			
2	Discount factor	1.00	0.91	0.83	
3	Year	0	1	2	TOTAL
4	Project 1				
5	Costs (cash outflows)	\$1,071,850	\$468,120	\$384,700	
6	Benefits (cash inflows)	\$0	\$1,250,000	\$1,750,000	
7	Net cash flow	(\$1,071,850)	\$781,880	\$1,365,300	
8					
9	Discounted costs	\$1,071,850	\$425,564	\$317,934	\$1,815,348
10	Discounted benefits	\$0	\$1,136,364	\$1,446,281	\$2,582,645
11	Discounted cash flow	(\$1,071,850)	\$710,800	\$1,128,347	\$767,297
12	Cumulative disc cash flow	(\$1,071,850)	(\$361,050)	\$767,297	
13	NPV	\$767,297			
14	ROI	42.27%			

- Assumptions for requirement of Staffing needs:

16	Assumptions					
17	Role	Number of Staff	Hourly Rate (AUD)	Estimated Hours per employee	Duration (months)	Total Cost (AUD)
18	Project Manager	1	\$75.00	1500	12	\$112,500.00
19	Game Developer	3	\$41.00	1470	12	\$180,810.00
20	Game Designer	2	\$38.00	1400	10	\$106,400.00
21	Animator	1	\$51.00	2720	8	\$138,720.00
22	UI/UX Designer	1	\$58.00	950	4	\$55,100.00
23	Data Scientist	1	\$62.00	1460	6	\$90,520.00
24	QA/Test Analyst	1	\$115.00	720	4	\$82,800.00
25	Total Staffing Cost:			10220		\$766,850.00
26	*The hourly cost of all the roles are taken from au.talent.com and glassdoor.com *					

- One-time hardware requirements:

31	EXHIBIT B - Hardware Device Cost		
32	Item	Estimated Cost (AUD)	Source/Assumption
33	Development Tools & Licenses	\$50,000.00	Licensing costs for tools like Unity Pro, Adobe Creative Cloud, and GitHub Enterprise .
34	Hardware Upgrades	\$20,000.00	Cost for upgrading workstations, including GPU and RAM upgrades .
35	Testing Hardware Devices	\$5,000.00	Cost of a buying consoles like PS4, XBOX, PC , Nintendo, etc.
36	Cloud Hosting (12 months)	\$30,000.00	Cost estimate for cloud hosting services like AWS.
37	Total Software/Hardware Cost:	\$105,000.00	
38	*The cost of AWS and google cloud is compared for data storage server and ASW is chosen*		
39	*The cost of hardware upgrades and testing devices have been taken from JB-Hi Fi and newegg*		
40	*The cost of licensing is taken from unreal engine and adobe creative plans*		

- Contingency/Opportunity Funds requirements:

48	Contingency Fund:	
49	Item	Estimated Cost (AUD)
50	Unforeseen Expenses	\$200,000.00
51	Total Contingency Fund:	\$200,000.00

- Year 1 and 2 Maintenance Cost requirements:

53	Maintenance Costs					
54	Year	Activity	Estimated Cost (AUD)			
55	1	Minor Updates and Bug Fixes	\$150,000.00			
56	2	Major Updates, Additional Content	\$200,000.00			
57	Total Maintenance Cost:		\$350,000.00			
58						
59						
60	Requirements for Year 0					
61	Role	Number of Staff	Hourly Rate (AUD)	Estimated Hours per employee	Duration (months)	Total Cost (AUD)
62	Project Manager	1	\$75.00	800	6	\$60,000.00
63	Game Developer	2	\$41.00	720	6	\$59,040.00
64	Data Scientist	1	\$62.00	540	6	\$33,480.00
65	QA/Test Analyst	2	\$115.00	720	4	\$165,600.00
66	Total Staffing Cost:			2780		\$318,120.00
67						
68	Requirements for Year 0					
69	Role	Number of Staff	Hourly Rate (AUD)	Estimated Hours per employee	Duration (months)	Total Cost (AUD)
70	Project Manager	1	\$75.00	740	6	\$55,500.00
71	Game Developer	2	\$41.00	560	6	\$45,920.00
72	Data Scientist	1	\$62.00	490	4	\$30,380.00
73	QA/Test Analyst	1	\$115.00	460	4	\$52,900.00
74	Total Staffing Cost:			2250		\$184,700.00

- Projected Benefits of the Project:

76	Benefits			
77	Year	Projected Revenue (AUD)	Justification	
78	1	\$1,250,000.00	Initial revenue driven by the game's launch, fueled by pre-launch marketing efforts, player interest in the genre, and positive initial reviews and feedback.	
79	2	\$1,750,000.00	Increased revenue due to the release of major updates and additional content, which will attract new players and encourage existing players to continue investing time.	
80	Total	\$3,000,000.00		

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